

RESERVE NETWORK STUDY

One network across the whole reserve.

We modelled Kwandwe's terrain from satellite elevation data and checked every radio path against it. Two ridge high-sites and one short relay put 15 of the 15 sites we could locate on clear radio paths and give line of sight across 53% of Kwandwe, fed from a carrier uplink on mapped mast, operator not tagged.

Prepared for

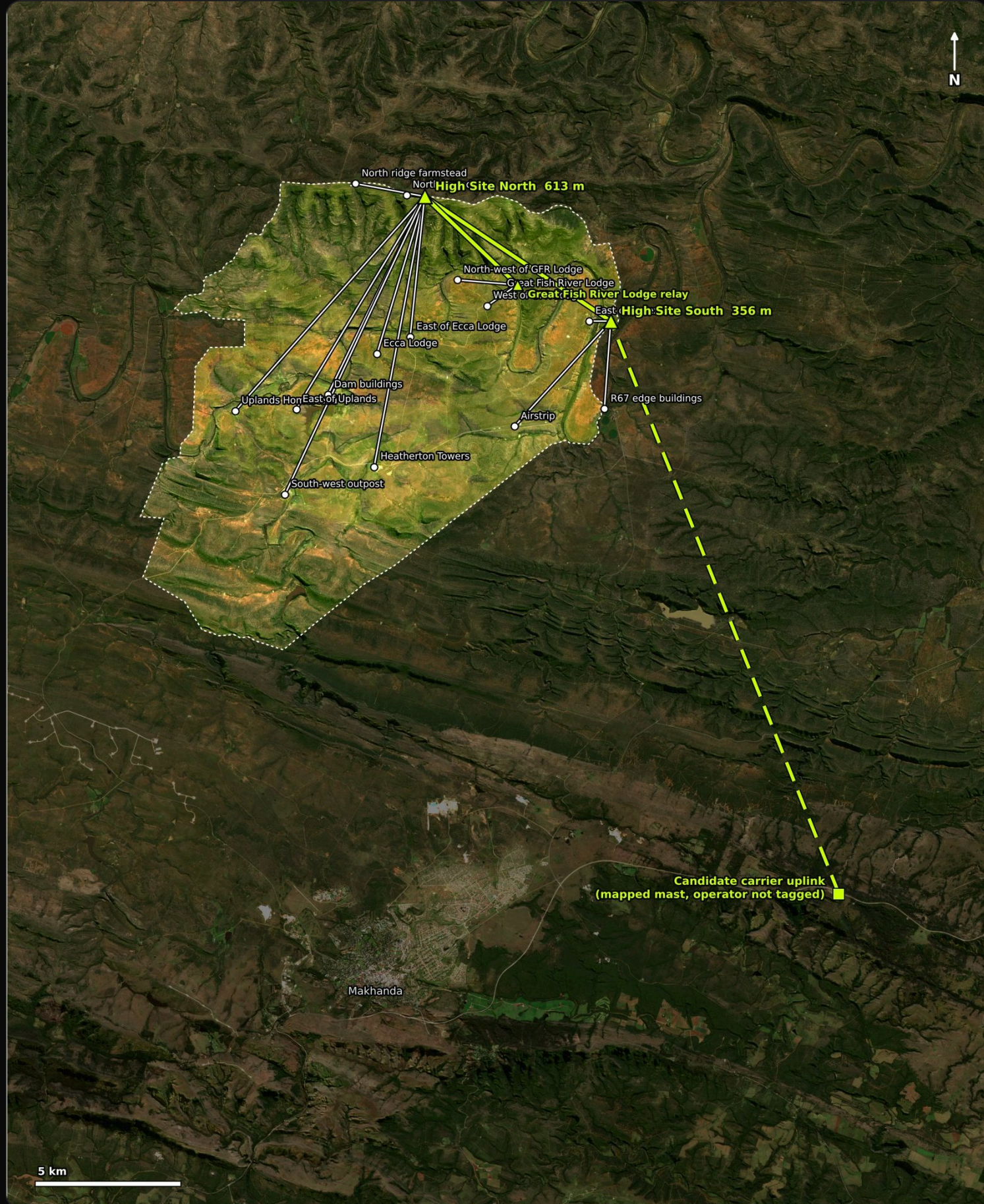
Angus Sholto-Douglas, Kwandwe Private Game Reserve

Prepared by

Gerhard Smit, CTTX Services (Pty) Ltd

Date

28 September 2026



--- Carrier uplink
 — Private backbone
 — Site link
■ Clear line of sight for a 3 m field terminal

Sentinel-2 imagery, 14 July 2026. The dotted outline is the mapped boundary of Kwandwe, 18,482 ha is mapped in OpenStreetMap. Published figures for the reserve run from 22,000 to 30,000 ha, so part of the property may lie outside this outline and is not modelled. Site positions are desktop candidates, to be confirmed on a site visit.

15 of 15

sites on a clear,
Fresnel-checked radio
path

2 + 1

ridge high-sites plus
short relays

53%

of the 185 km² study
area in line of sight

22.0 km

clear carrier uplink into
the reserve

01

Where things stand today

Kwandwe runs four lodges, a security operation and an environmental team across a property split by the Great Fish River valley. Lodges on the valley side rarely see each other directly. One owned backbone on the ridges can reach all of them, the gates and the field, on a network the reserve controls.

WHAT THE PUBLIC RECORD SHOWS

- Four lodges (Great Fish River Lodge, Ecca Lodge, Melton Manor, Uplands Homestead), Heatherton Towers, an airstrip and at least ten other building groups on one reserve
- The Great Fish River runs about 30 km through the property and almost divides it in half
- Security (TFT Security) and the environmental team are already part of this conversation
- No reserve-wide network is described for rangers, gates or cameras

02

Carrier in, private network out

Carrier connectivity comes into the reserve once. Everything past that point runs on infrastructure the reserve owns.

LAYER 1 · CARRIER

Uplink from mapped mast, operator not tagged

An uncontended business-grade Vodacom service, delivered by CTTX over a 22.0 km link to High Site South. LTE failover keeps the reserve connected if the carrier path drops.

LAYER 2 · BACKBONE

Ridge high-sites

High Site South (356 m) and High Site North (613 m) sit 8.0 km apart with a clear link between them. A short Great Fish River Lodge relay serves a lodge the ridges cannot see. Each site runs on solar and lithium storage with remote monitoring, so the network stays up through load-shedding.

LAYER 3 · DISTRIBUTION

Every lodge and ops site

Sector radios reach the lodges, HQ, the anti-poaching base, ecology, the workshop and the gates. A LoRaWAN gateway is ready for fence, water-point, pump and fuel sensors.

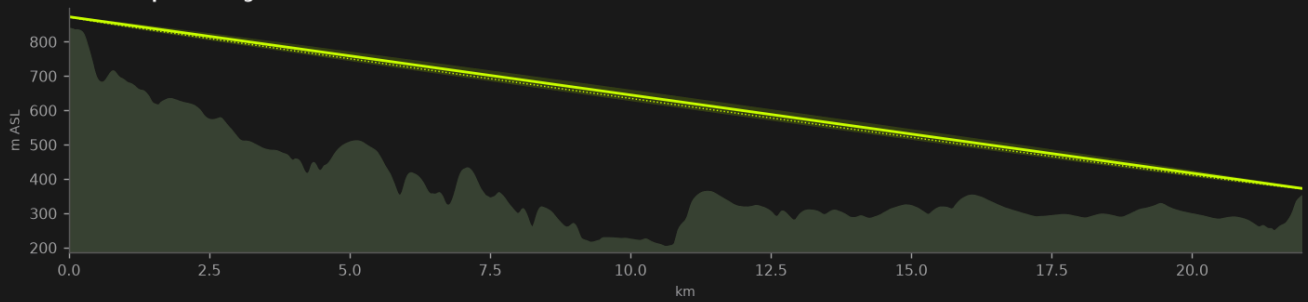
BUILT FOR WHAT COMES NEXT

Once the backbone covers the reserve, a new camera, sensor or lodge is an endpoint on a network that is already there. It is not a new infrastructure project or a new provider contract.

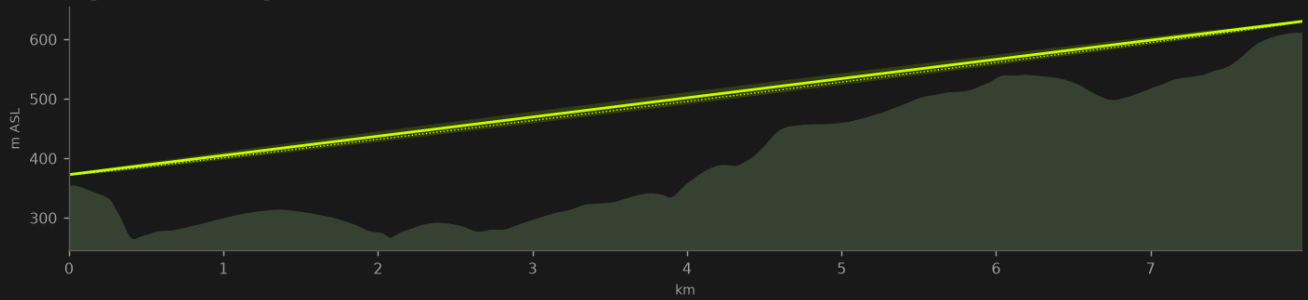
03 Every path checked against the terrain

Each profile shows the ground between two sites, the radio path (lime line) and its Fresnel zone (lime band). We allow for earth curvature and require at least 60% of the zone to be clear at every point. A path that fails is not drawn.

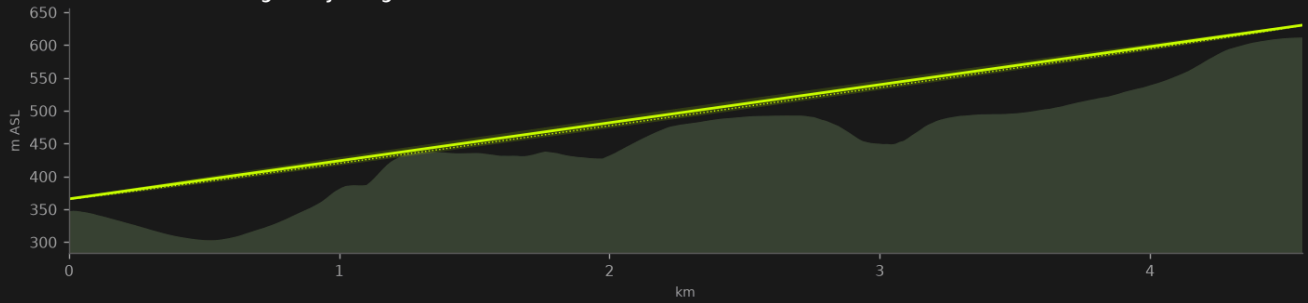
Carrier uplink → High Site South 22.0 km · Fresnel zone clear



High Site South → High Site North 8.0 km · Fresnel zone clear

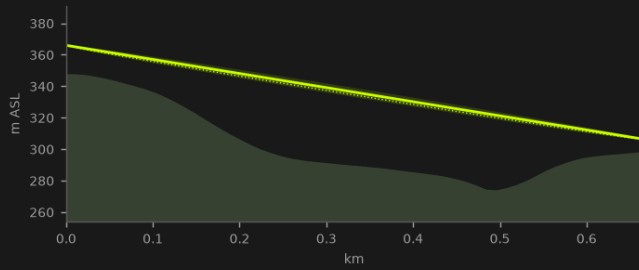


Great Fish River Lodge relay → High Site North 4.6 km · Fresnel zone clear

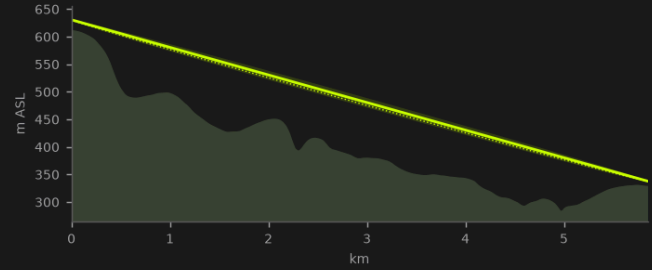


SITE	SERVED FROM	PATH LENGTH	LINE OF SIGHT
Great Fish River Lodge	Great Fish River Lodge relay	0.7 km	CLEAR
Ecca Lodge	High Site North	5.9 km	CLEAR
Uplands Homestead	High Site North	10.2 km	CLEAR
Heatherton Towers	High Site North	9.8 km	CLEAR
Airstrip	High Site South	5.1 km	CLEAR
East complex	High Site South	0.8 km	CLEAR
West of GFR Lodge	Great Fish River Lodge relay	1.3 km	CLEAR
East of Ecca Lodge	High Site North	5.0 km	CLEAR
North-west of GFR Lodge	Great Fish River Lodge relay	2.1 km	CLEAR
Dam buildings	High Site North	7.9 km	CLEAR
East of Uplands	High Site North	8.9 km	CLEAR
South-west outpost	High Site North	11.7 km	CLEAR
North boundary	High Site North	0.7 km	CLEAR
North ridge farmstead	High Site North	2.5 km	CLEAR
R67 edge buildings	High Site South	3.1 km	CLEAR
Melton Manor	—	—	POSITION TO CONFIRM

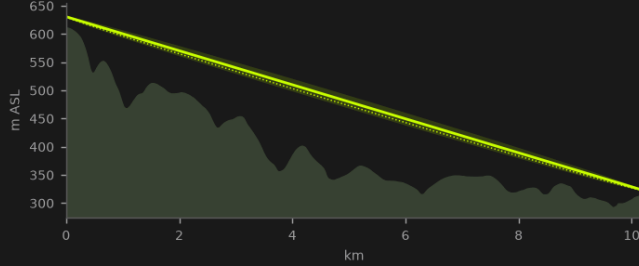
Great Fish River Lodge relay → Great Fish River Lodge 0.7 km · Fresnel zone clear High Site North → Ecce Lodge 5.9 km · Fresnel zone clear



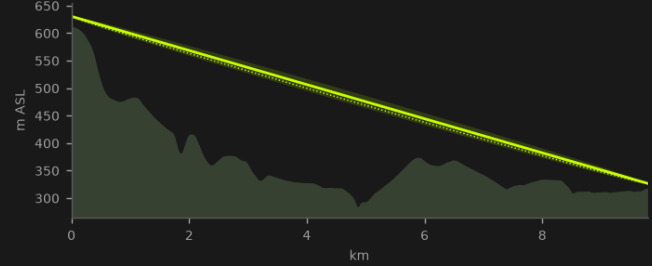
High Site North → Eccca Lodge 5.9 km · Fresnel zone clear



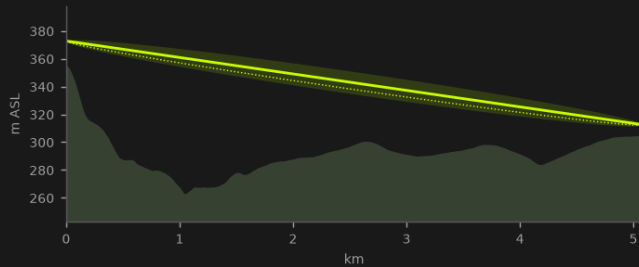
High Site North → Uplands Homestead 10.2 km · Fresnel zone clear



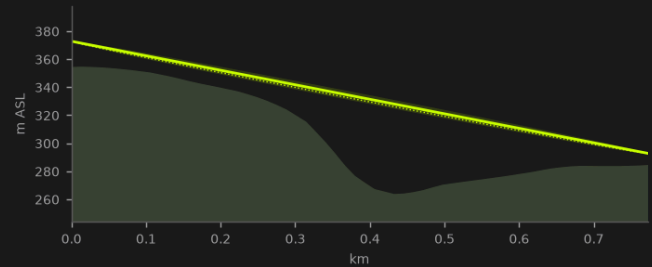
High Site North → Heatherton Towers 9.8 km · Fresnel zone clear



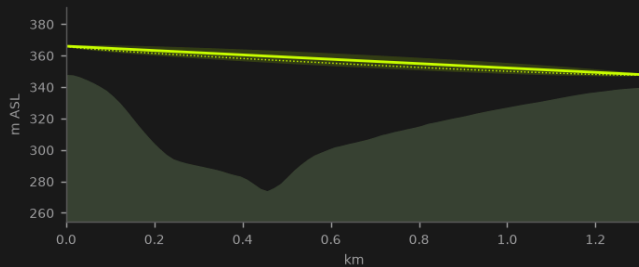
High Site South → Airstrip 5.1 km · Fresnel zone clear



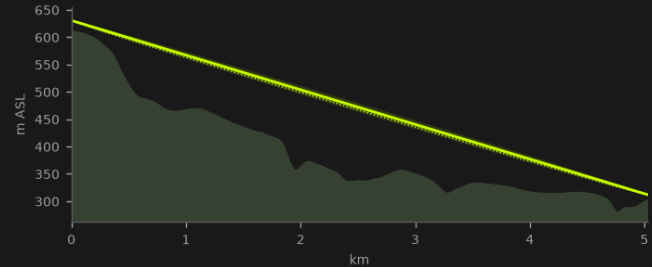
High Site South → East complex 0.8 km · Fresnel zone clear



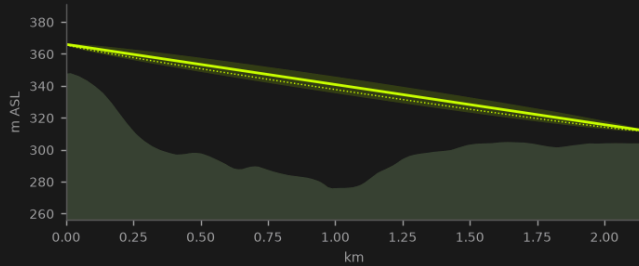
Great Fish River Lodge relay → West of GFR Lodge 1.3 km · Fresnel zone clear



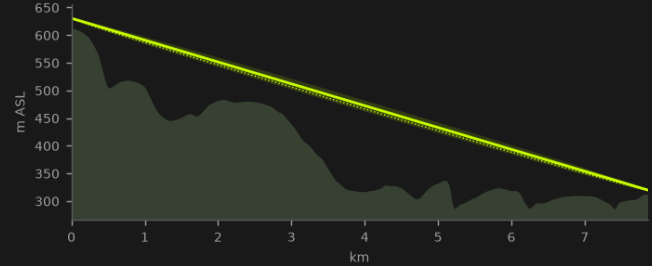
High Site North → East of Eccla Lodge 5.0 km · Fresnel zone clear



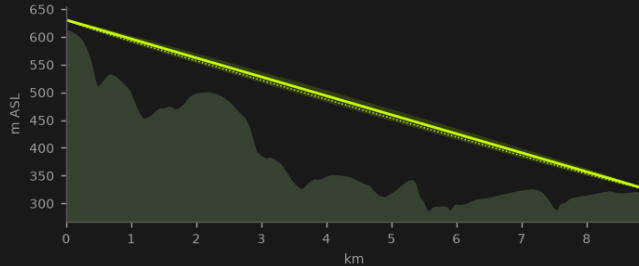
Great Fish River Lodge relay → North-west of GFR Lodge 2.1 km · Fresnel zone clear



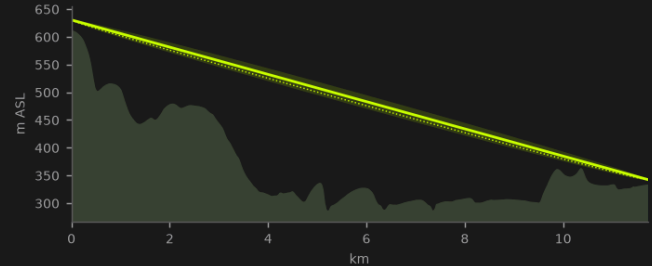
ed High Site North → Dam buildings 7.9 km · Fresnel zone clear



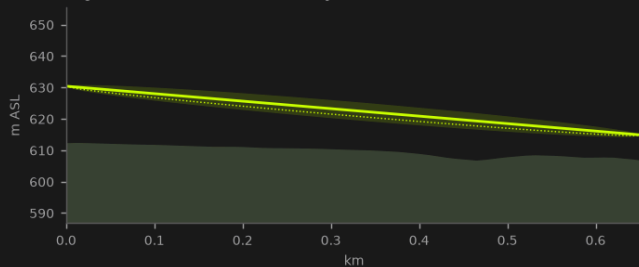
High Site North → East of Uplands 8.9 km · Fresnel zone clear



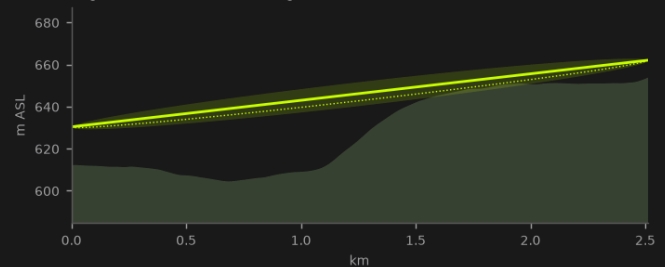
High Site North → South-west outpost 11.7 km · Fresnel zone clear



High Site North → North boundary 0.7 km · Fresnel zone clear

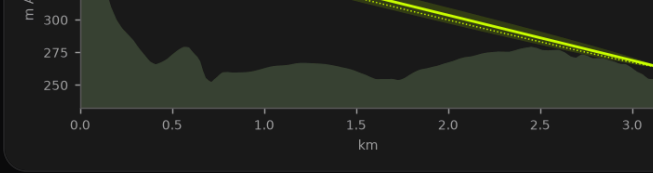


High Site North → North ridge farmstead 2.5 km · Fresnel zone clear



High Site South → R67 edge buildings 3.1 km · Fresnel zone clear





Entries marked “position to confirm” have no published location. We add them to the model once you share a marked map or KMZ.

5.8 GHz · $k = 4/3$ · uplink tower 30 m · high-site masts 18 m · site masts 8 m. Site positions come from mapped buildings and published places (OpenStreetMap, Google/Microsoft building footprints via Overture Maps).

04 One backbone for every driver of the business

A reserve network carries three loads at once. This design is built to hold all three on one backbone.

GUEST EXPERIENCE

Reliable Wi-Fi, booking and card payments at every lodge.

STAFF COMMUNICATION

Lodges, HQ, rangers and maintenance on one network, day and night.

SECURITY AND OPERATIONS

Cameras, gates, fence alarms and sensors on the same backbone, powered through load-shedding.

05 What other properties got back

Sandymount Safaris 30,000+ ha, three lodges

BEFORE

A consumer wireless service at about 3 Mbps down and 8 Mbps up, with the provider on site daily to swap IPs and reinstall equipment.

AFTER

A Vodacom-backed 100 Mbps symmetrical link into the reserve, a CTTX-built main mast and two secondary masts, solar and lithium power, all monitored remotely.

GUEST EXPERIENCE

HD sport streams live at the lodges. Guest connectivity is no longer something the lodges think about.

STAFF COMMUNICATION

Over 800 Mbps measured across the reserve's own backbone, linking the main lodge, Mountain Retreat and Samar.

SECURITY AND OPERATIONS

Daily provider callouts went to zero. Every fault used to need a vehicle and a person; now the network is monitored remotely.

OWNERSHIP

The reserve keeps adding systems on top of the backbone. In the client's words, they stopped doing speed tests.

WHEN THE NETWORK GOES QUIET AFTER HOURS

Elephants broke out after 21:00 and staff had no effective way to coordinate. The requirement that came out of it: dependable communications after hours, across the areas where an incident can happen.

An Eastern Cape lodge near Addo

OWNING THE NETWORK PAYS FOR ITSELF

Replacing a rented, contended service and rented Wi-Fi with a lodge-owned network breaks even in under three years on connectivity savings alone. It also connects areas the old setup never reached, including the spa.

The same lodge, business case from its own invoices (Sep 2026)

06

What it pays back

A reserve that replaces a separate rented connection and rented Wi-Fi at every lodge with one owned backbone typically recovers the investment in under three years on connectivity costs alone. The security, operations and guest-experience benefits come on top of that.

YOUR NUMBER, NOT OURS

Send us each lodge's latest connectivity invoice, or just the monthly totals, and we will work out Kwandwe's actual payback. It takes us a day.

If infrastructure decisions sit with someone else, please pass this on. Owners will find the payback and ownership case, the reserve manager the security and operations case, and lodge managers the guest experience case.

07

What changes when the reserve owns it

TODAY

- Four lodges (Great Fish River Lodge, Ecca Lodge, Melton Manor, Uplands Homestead), Heatherton Towers, an airstrip and at least ten other building groups on one reserve
- The Great Fish River runs about 30 km through the property and almost divides it in half
- Security (TFT Security) and the environmental team are already part of this conversation
- No reserve-wide network is described for rangers, gates or cameras

WITH A RESERVE BACKBONE

- One uncontended carrier feed shared across the reserve
- Lodges, HQ, the anti-poaching base and gates on one private network
- Coverage out into the field, day and night
- An owned asset that takes cameras and sensors without new towers

Avoided loss

Kwandwe is a Big Five reserve with rhino, and its security contractor is part of this discussion. A backbone that reaches the lodges, gates and ranger areas carries cameras, radio and alarms on one monitored network, day and night.

Operational efficiency

Water points, pumps and fuel tanks report their own status, so there are fewer vehicle trips just to check. Maintenance, ecology and lodges share one communications path.

Infrastructure ownership

High sites, backbone and power belong to the reserve. The network grows with the reserve instead of depending on each lodge's separate provider.

Executive risk reduction

Management gets live visibility of the network, cameras and sensors, and a record of operational diligence for owners, insurers and conservation partners.

Next steps

1 30-minute call

Your lodges' current connectivity costs, your peak usage, the incidents that were hardest to coordinate, and what currently needs a vehicle trip to check.

2 Confirm the sites

A marked map of HQ, the anti-poaching base and the gates. We add them to the model and rerun the link plan.

3 One day on the reserve

We walk the high sites and relay points, and confirm the carrier uplink with Vodacom.

4 Design and proposal

A final design in Cambium LINKPlanner, with equipment, power, a phased plan and the investment case.

Method. Terrain: Copernicus DEM GLO-30 (30 m). Imagery: ESA Sentinel-2 L2A true colour, 14 July 2026. Radio model: 5.8 GHz, effective earth radius factor 4/3, 60% first-Fresnel-zone clearance at every point. The terrain model does not include trees or buildings. Every path is confirmed on survey before anything is built.

Candidate positions. High Site North -33.05981, 26.54515; High Site South -33.10010, 26.61599; Great Fish River Lodge relay -33.08835, 26.58036; carrier uplink -33.28428, 26.70253 (mapped mast, operator not tagged). The exact carrier site is subject to Vodacom feasibility.

Gerhard Smit

CTTX Services (Pty) Ltd

Tel 084 550 3281

Email gerhard@cttx.co.za

Web cttx.co.za